# Vanishing of the point-to-point exponent in high dimension

An endpoint-attachment argument for the Ding–Gwynne–Zhuang white-noise field

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**Abstract**

For the unit-ball white-noise regularization, the weighted second-moment construction in Ding–Gwynne–Zhuang can be combined with Gaussian association to attach prescribed endpoints. For every $\alpha > 0$, in all sufficiently large dimensions, the condition $\psi\alpha > \log(11/4)$ gives

$$2^{-n}e^{-C\sqrt{n}} \le \operatorname{Med} D_n(0, e_1) \le \operatorname{Med} D_n(0, e_1; (-2, 2)^d) \le 2^{-n}e^{C\sqrt{n}}.$$

In particular $Q_d(\psi) = 0$. The endpoint step uses a weighted count rather than an estimate at an endpoint selected by the field. Neither existence of the set-to-set exponent nor a comparison between the two exponents is assumed.

**Provenance.** This note gives a new argument beyond the statements of [1]. Its imported input is the weighted bridge construction in the proof of Proposition 3.19 of that paper, including its second-moment estimate. The extraction of that input is specified in Section 3; the remaining steps are proved here. This new argument has not undergone independent expert review.

## 1 Setup and result

Let $W$ be space-time white noise on $\mathbb{R}^d \times (0, \infty)$, and set

$$K(z) = \operatorname{vol}(B_1(0))^{-1/2} \mathbf{1}_{\{|z|_2 < 1\}}.$$

For real $a < b$, define

$$H_{a,b}(x) = \int_{2^{-b}}^{2^{-a}} \int_{\mathbb{R}^d} K\left(\frac{y - x}{t}\right) t^{-(d+1)/2} W(dy, dt), \qquad h_n = H_{0,n}, \quad h_0 = 0. \tag{1.1}$$

We use continuous versions. For $\psi > 0$ and an open set $U$, let

$$D_n(x, y; U) = \inf_{P: x \to y,\, P \subset U} \int_0^1 e^{\psi h_n(P(t))} |P'(t)|_2 \, dt,$$

where paths are piecewise continuously differentiable. Omit $U$ when it is $\mathbb{R}^d$. Throughout, Med is the lower 1/2-quantile and logarithms are natural. Write

$$\mathcal{C} = (-2, 2)^d, \qquad a_n = \operatorname{Med} D_n(0, e_1; \mathcal{C}). \tag{1.2}$$

The parameter $\psi$ is denoted by $\xi$ in [1]. The parameter $\alpha$ below is normalized as in Proposition 3.19 of that paper: the field bound is $\alpha(j - 3k)$, without an additional factor of $\sqrt{\log 2}$.

**Theorem 1.1.** *For every $\alpha > 0$, there is $d_*(\alpha) < \infty$ such that, whenever*

$$d \geq d_*(\alpha), \qquad \psi > 0, \qquad \psi\alpha > \log(11/4), \tag{1.3}$$

*there is $C = C(d, \alpha, \psi) < \infty$ with*

$$2^{-n}e^{-C\sqrt{n}} \leq \operatorname{Med} D_n(0, e_1) \leq a_n \leq 2^{-n}e^{C\sqrt{n}} \qquad (n \geq 1). \tag{1.4}$$

*Consequently the point-to-point exponent in [1, Equation (1.8)] satisfies $Q_d(\psi) = 0$. Every fixed quantile has the same exponential rate, and for each fixed $x \neq y$,*

$$\frac{\log D_n(x, y)}{n \log 2} \longrightarrow -1 \quad \textit{almost surely.} \tag{1.5}$$

**Corollary 1.2.** *For every $\psi_0 > 0$, there exists $d_0(\psi_0) < \infty$ such that*

$$d \geq d_0(\psi_0), \quad \psi \geq \psi_0 \quad \Longrightarrow \quad Q_d(\psi) = 0.$$

*Proof.* Choose any $\alpha > \log(11/4)/\psi_0$ and apply Theorem 1.1 with $d_0(\psi_0) = d_*(\alpha)$. Constants in (1.4) need not be uniform over $\psi \geq \psi_0$. □

## 2 Gaussian estimates and deterministic endpoints

All constants in this section may depend on $d, \psi$, a fixed bounded region, and a fixed probability of failure, but not on scale indices or on the deterministic endpoints over the ranges specified.

**Lemma 2.1** (Concentration)**.** *Let $X_n$ be a point-to-point distance, internal to a fixed bounded open connected set or unrestricted. Then*

$$\mathbb{P}\big[|\log X_n - \log \operatorname{Med} X_n| > t\big] \leq 2\exp\left(-\frac{t^2}{2\psi^2 n \log 2}\right), \qquad t > 0. \tag{2.1}$$

*The same statement holds for distances between disjoint compact endpoint sets for which a finite polygonal connection exists.*

*Proof.* A Cameron–Martin shift $u \in L^2(\mathbb{R}^d \times (0, \infty))$ changes $h_n(x)$ by at most $\sqrt{n \log 2}\, \|u\|_2$, simultaneously for all $x$. Comparing the length of every path shows that $\log X_n$ is $\psi\sqrt{n \log 2}$-Lipschitz in the Cameron–Martin norm. Gaussian isoperimetry gives (2.1). Positivity and integrability of its logarithm follow by forcing each path to leave a fixed neighborhood of its first endpoint, and by bounding above with a fixed polygonal path. Measurability follows from a countable polygonal approximation and continuity of the field. In particular, passing between any two fixed quantiles changes logarithmic distance by at most $C\sqrt{n}$. □

**Lemma 2.2** (Field bounds)**.** *The following properties hold.*

*(i) For $a < b$,*

$$\operatorname{Var} H_{a,b}(x) = (b - a)\log 2, \qquad \mathbb{E}|H_{a,b}(x) - H_{a,b}(y)|^2 \leq C_d 2^b |x - y|_2. \tag{2.2}$$

*(ii) For each $R < \infty$ and $\delta > 0$, there is $C_{R,\delta} < \infty$ such that, uniformly in $m \geq 0$ and deterministic $x$,*

$$\mathbb{P}\left[\sup_{|z-x|_2 \leq R2^{-m}} |h_m(z)| \leq C_{R,\delta}(1 + \sqrt{m})\right] \geq 1 - \delta. \tag{2.3}$$

*(iii) There is $B = B(d) < \infty$ such that*

$$\mathbb{P}\left[\sup_{z\in\mathcal{C}} h_s(z) \le Bs\right] \ge 0.99 \quad \textit{for all integers } s \ge 1. \tag{2.4}$$

*(iv) For each $N \ge 1$, $s \in [1/2, 2]$, and orthogonal $R$, the field $u \mapsto h_N(sRu)$ can be coupled with a copy of $h_N(u)$ so that*

$$\mathbb{P}\left[\sup_{u\in\mathcal{C}} |\widehat{h}_N(u) - h_N(u)| \le C_\delta\sqrt{N}\right] \ge 1 - \delta. \tag{2.5}$$

*The constant is uniform over deterministic $s, R$; the assertion does not require a simultaneous coupling over all $s, R$.*

*Proof.* White-noise isometry and $\int K^2 = 1$ give the variance. The symmetric difference of two translated unit balls has volume at most $C_d \min\{|v|_2, 1\}$. Thus

$$\|K(\cdot - v) - K\|_2^2 \le C_d \min\{|v|_2, 1\},$$

and integrating $C_d|x - y|_2 t^{-2}$ from $2^{-b}$ to $2^{-a}$ proves (2.2). Gaussian moments of arbitrary order give a continuous version by Kolmogorov's criterion.

On the rescaled ball $|u|_2 \le R$, the centered process $h_m(x + 2^{-m}u) - h_m(x)$ has canonical metric at most $C_d|u - v|_2^{1/2}$ and uniformly bounded pointwise variance. Its covering numbers at canonical radius $\varepsilon$ are bounded by $C_{d,R}\varepsilon^{-2d}$. The Gaussian entropy integral is finite, so its expected absolute supremum is bounded independently of $m, x$. Gaussian supremum concentration gives an $O_{d,R,\delta}(1)$ bound with probability $1 - \delta/2$. Combining this with $h_m(x) \sim N(0, m \log 2)$ proves (ii).

For (iii), cover $\overline{\mathcal{C}}$ by $O_d(2^{ds})$ cubes of side $2^{-s}$. The maximum of their center values has expectation $O_d(s)$, by the Gaussian tail bound with variance $s \log 2$. The oscillation on each cube has bounded expected supremum and a uniform Gaussian tail, as above. The maximum of these oscillations has expectation $O_d(\sqrt{s})$. Consequently $\mathbb{E}\sup_{\mathcal{C}} h_s \le C_d s$. Gaussian supremum concentration, with variance parameter $s \log 2$, gives (iii) after increasing $B$.

For (iv), radial symmetry and white-noise scaling give, with $c = \log_2 s \in [-1, 1]$,

$$\{h_N(sRu) : u \in \mathcal{C}\} \overset{\text{law}}{=} \{H_{c,N+c}(u) : u \in \mathcal{C}\}.$$

Couple this with $H_{0,N}$ using the same white noise. For $c \ge 0$ the difference is $H_{N,N+c} - H_{0,c}$; for $c < 0$ it is $H_{c,0} - H_{N+c,N}$. The coarse band has an absolute supremum bounded in probability, uniformly over $c \in [-1, 1]$. The fine band has width at most one. Covering $\mathcal{C}$ by $O_d(2^{dN})$ microscopic cubes, its pointwise variance and its rescaled local oscillations are uniformly bounded. The Gaussian union bound therefore bounds its absolute supremum by $C_\delta\sqrt{N}$ with probability $1 - \delta$. This proves (iv). □

**Lemma 2.3** (Attachment to a deterministic nearby point)**.** *Fix $\delta > 0$. For integers $n = m + N$ with $m \ge 0, N \ge 1$, a deterministic $x \in \mathbb{R}^d$, and a deterministic $v$ with $1/2 \le |v|_2 \le 2$, choose an orthogonal map $R_v$ satisfying $R_v e_1 = v/|v|_2$, and let*

$$U_{x,m,v} = x + 2^{-m}|v|_2 R_v \mathcal{C}.$$

*There is $C_\delta < \infty$, uniform in these choices, such that*

$$\mathbb{P}\Big[D_n(x, x + 2^{-m}v; U_{x,m,v}) \le 2^{-m} e^{C_\delta\sqrt{n}} a_N\Big] \ge 1 - \delta. \tag{2.6}$$

*Moreover $U_{x,m,v} \subset B_{4\sqrt{d}\,2^{-m}}(x)$.*

*Proof.* Decompose $h_n = h_m + H_{m,n}$. With $\rho = 2^{-m}$, the field $u \mapsto H_{m,n}(x + \rho u)$ has the law of $h_N(u)$. Changing variables in the fine-field metric gives a factor $\rho|v|_2$, with field $h_N(|v|_2 R_v u)$ on $\mathcal{C}$. By Lemma 2.2(iv) and Lemma 2.1, this fine-field internal distance is at most

$$\rho a_N \exp(C_\delta \sqrt{N})$$

with probability at least $1-\delta/2$, uniformly in $v$. The factor $|v|_2 \le 2$ is absorbed into the exponential. The whole domain is contained in the indicated ball, so Lemma 2.2(ii) bounds the coarse field there by $C_\delta(1+\sqrt{m})$ with probability $1-\delta/2$. Multiplying path lengths by this coarse density and using $\sqrt{m}+\sqrt{N} \le \sqrt{2n}$ proves (2.6). Only a union bound is used. □

*Remark* 2.4. Lemma 2.3 is a uniform *marginal* estimate for deterministic endpoints. It is not asserted at an endpoint selected from the field. The next two sections provide the step that legitimately permits a field-dependent path choice.

## 3 The weighted bridge input

The needed input is stronger than the existential statement of Proposition 3.19 in [1]: we retain the weighted count used in its proof. Here and below a path denotes its polygonal interpolation when field values or Euclidean length are considered.

**Lemma 3.1** (Weighted bridges from the DGZ construction)**.** *Fix $\alpha > 0$. For all sufficiently large $d$, one can choose an even integer $k = k(d,\alpha)$ such that the following holds for every $N \ge 6k$ and every integer $M \ge 10$. There is a finite deterministic family $\mathcal{F}_{N,M}$ of paths, with associated events $G_P$ of positive probability, having these properties:*

*(i) Each path has endpoints $p_P, q_P$ satisfying*

$$\tfrac{1}{2} \le |p_P|_2 \le 2, \qquad \tfrac{1}{2} \le |q_P - 2Me_1|_2 \le 2.$$

*Every path is contained in*

$$[-1, 2M+1] \times [-1, M+1]^{d-1}$$

*and has Euclidean length at most $4M(11/8)^N$.*

*(ii) On $G_P$,*

$$h_N(z) - h_{3k}(z) \le -\alpha(N-3k) \qquad (z \in P). \tag{3.1}$$

*(iii) Each $G_P$ is decreasing in Gaussian variables obtained by integrating nonnegative deterministic kernels against white noise.*

*(iv) With $J = |\mathcal{F}_{N,M}|$ and $\pi_P = \mathbb{P}(G_P)$,*

$$Z = \frac{1}{J} \sum_{P \in \mathcal{F}_{N,M}} \frac{\mathbf{1}_{G_P}}{\pi_P} \quad \text{satisfies} \quad \mathbb{E}Z = 1, \qquad \mathbb{E}Z^2 \le \frac{5}{4}. \tag{3.2}$$

*The choice of $k$ and the moment bound are uniform in both $M$ and $N$.*

*Extraction from [1].* Use the bridge family in the proof of Proposition 3.19, with $\mathfrak{r} = \lfloor \sqrt{d} \rfloor$. Its unrefined paths start at $e_{i_1}$ and end at $2Me_1 + e_{i_1}$, for sequences $(i_1, \dots, i_M) \in \{2, \dots, d\}^M$. Their first coordinate is between 0 and $2M$; the other coordinates are between 0 and $M$. The refinement

procedure of Section 3.2 stays within $2/(7\mathfrak{r})$ in $\ell^1$ distance of the base path. Each endpoint moves by at most the same amount, by summing its successive finer-grid displacements. The number of edges is at most $4M\mathfrak{r}\,11^N$, each of Euclidean length $\mathfrak{r}^{-1}8^{-N}$. These facts give (i).

Use the good events of Definition 3.13 for this bridge family and reflect all white noise by $W \mapsto -W$. Their constraints are upper bounds on the auxiliary integrals in (3.14), and on $H_{3k,j}(y)$ in (3.15). All these are integrals of nonnegative kernels. Remark 3.14 gives the bound on the entire interpolated path in (ii). This proves (ii)–(iii).

For completeness, the precise second-moment extraction is as follows. For two independent uniform paths let $Y$ count intersections of the unrefined bridges and let $Y_j$ count intersections after interpolation and $j$ refinements. The normalized count in (3.2) is the count in (3.16) divided by its expectation. Equations (3.43)–(3.46), which the proof of Proposition 3.19 applies to the bridge family, give, for large even $k$,

$$\mathbb{E}Z^2 \le \mathbb{E}_{P,Q}\Big[(1-2e^{-k})^{-6Y}(1+2^{-k/2})^{Y_0}\Big]. \tag{3.3}$$

Here $Y_0 \le 2\mathfrak{r}Y$. The bridge comparison in the proof of Proposition 3.19 stochastically bounds $Y$ by $4G$, where $G$ has a geometric tail depending on $d$ but not on $M$. Therefore, with

$$c_k = (1-2e^{-k})^{-6}(1+2^{-k/2})^{2\mathfrak{r}},$$

we obtain

$$\mathbb{E}Z^2 \le \mathbb{E}[c_k^{4G}] \longrightarrow 1 \qquad (k \longrightarrow \infty).$$

The convergence is uniform in $M$ and $N \ge 6k$, because $G$ has a fixed exponential moment and $c_k \to 1$. Choosing $k$ once and for all gives (iv). The identity $\mathbb{E}Z = 1$ is immediate from the normalization. This uses the moment proof itself, not an inference of a second-moment bound from the mere existence of a good path. □

# 4 Monotone thinning of the weighted count

A Gaussian vector whose covariance entries are all nonnegative is associated: two bounded coordinatewise decreasing functions of that vector have nonnegative covariance. One proof uses Gaussian interpolation between independent copies and identical copies. For smooth functions the derivative is a sum of terms

$$\mathrm{Cov}(X_i, X_j)\,\mathbb{E}[\partial_i f(X)\partial_j g(Y_t)] \ge 0.$$

Integration and approximation yield the assertion for events. It extends to the continuous Gaussian fields here: use a countable dense set for field inequalities and finite piecewise-affine interpolation for metric functionals. Interpolation with nonnegative barycentric coefficients preserves monotonicity, and uniform convergence of the field on a compact set gives convergence of every internal distance there. Smooth one-sided approximations to event indicators handle possible equality at thresholds.

In this application all jointly considered Gaussian variables are white-noise integrals of nonnegative kernels. Their pairwise covariances are therefore nonnegative. In particular, a low-field good-path event, an upper bound on a metric distance, and an upper bound on the coarse field all have the same, decreasing, monotonicity.

**Lemma 4.1** (Weighted thinning)**.** *Let $G_1, \dots, G_J$ be decreasing events with $p_i = \mathbb{P}(G_i) > 0$, and suppose*

$$Z = \frac{1}{J}\sum_{i=1}^{J} \frac{\mathbf{1}_{G_i}}{p_i}, \qquad \mathbb{E}Z^2 \le K.$$

*For each deterministic index $i$, let $A_i$ be a decreasing event in the same associated Gaussian family, and suppose $\mathbb{P}(A_i) \geq p$. Then*

$$\mathbb{P}\left[\textit{there exists } i \textit{ with } G_i \cap A_i\right] \geq \frac{p^2}{K}. \tag{4.1}$$

*Proof.* Define

$$Z_A = \frac{1}{J}\sum_{i=1}^{J} \frac{\mathbf{1}_{G_i \cap A_i}}{p_i}.$$

Association gives $\mathbb{P}(G_i \cap A_i) \geq p_i \mathbb{P}(A_i) \geq p_i p$, hence $\mathbb{E}Z_A \geq p$. Also $0 \leq Z_A \leq Z$, so $\mathbb{E}Z_A^2 \leq K$. Cauchy–Schwarz gives

$$p^2 \leq (\mathbb{E}Z_A)^2 \leq \mathbb{P}(Z_A > 0)\mathbb{E}Z_A^2 \leq K\mathbb{P}(Z_A > 0),$$

which is (4.1). □

*Remark* 4.2 (Why this addresses the endpoint issue). The events $A_P$ will require cheap connections to the two endpoints of a *deterministic candidate* $P$. Lemma 2.3 controls each such event marginally. Lemma 4.1 then selects a candidate with both properties. There is no union bound over the candidate family, no estimate for the diameter of an endpoint ball, and no assertion of independence between the chosen path and its attachments.

## 5 The median recursion

Fix $\alpha, d, \psi$ as in (1.3), and fix $k$ from Lemma 3.1. Let $B$ be as in Lemma 2.2(iii), and set

$$\beta = \psi\alpha - \log(11/8), \qquad \kappa = \beta - \log 2 > 0, \qquad \theta = \frac{\kappa}{4(\beta + \psi B)} \in (0, 1/4). \tag{5.1}$$

For sufficiently large $n$ define

$$m = \lfloor \theta n \rfloor, \qquad N = n - m, \qquad \rho = 2^{-m}, \qquad M = 2^{m-1}. \tag{5.2}$$

Thus $M$ is an integer, $2M\rho = 1$, and, for large $n$, $M \geq 10$ and $N \geq 6k$.

Apply the weighted bridge construction to the rescaled fine white noise, whose associated fields are

$$F_j(u) = H_{m,m+j}(\rho u), \qquad j \geq 0.$$

The full family $(F_j)$ has the same law as $(h_j)$. It is convenient to define it also for $j > N$: the auxiliary good-path constraints can involve finer white noise than the cutoff of the metric. This is allowed on the original white-noise probability space and does not change $D_n$.

For $P \in \mathcal{F}_{N,M}$, write $\gamma_P = \rho P$, and denote its endpoints by $u_P, v_P$. Lemma 3.1 gives

$$\frac{\rho}{2} \leq |u_P|_2 \leq 2\rho, \qquad \frac{\rho}{2} \leq |v_P - e_1|_2 \leq 2\rho, \qquad \mathrm{len}_{\mathrm{Euc}}(\gamma_P) \leq 2(11/8)^N. \tag{5.3}$$

Moreover,

$$\gamma_P \subset [-\rho, 1+\rho] \times [-\rho, 1/2 + \rho]^{d-1} \subset \mathcal{C}.$$

All endpoint attachment domains in Lemma 2.3, centered at 0 and $e_1$, also lie in $\mathcal{C}$ once $4\sqrt{d}\,\rho < 1$.

Choose the constant in Lemma 2.3 with failure probability 0.005, and write

$$T_n = 2^{-m} e^{C\sqrt{n}} a_N. \tag{5.4}$$

For each deterministic candidate $P$, let $A_P$ be the intersection of the following three events:

(i) $D_n(0, u_P; U_{0,m,u_P/\rho}) \le T_n$;

(ii) $D_n(e_1, v_P; U_{e_1,m,(v_P-e_1)/\rho}) \le T_n$;

(iii) $\sup_{z\in\mathcal{C}} h_{m+3k}(z) \le B(m+3k)$.

For every deterministic $P$, the union bound gives

$$\mathbb{P}(A_P) \ge 1 - 0.005 - 0.005 - 0.01 = 0.98. \tag{5.5}$$

Each of these events is decreasing in the Gaussian variables described in Section 4. This includes the reflected fine-field good event $G_P$. Hence Lemma 4.1 and (3.2) give

$$\mathbb{P}[\exists P : G_P \cap A_P] \ge \frac{0.98^2}{5/4} > \frac{1}{2}. \tag{5.6}$$

The estimate is uniform in $M$, even though $M$ grows exponentially with $n$.

On $G_P \cap A_P$, for $z \in \gamma_P$,

$$h_n(z) = h_{m+3k}(z) + H_{m+3k,n}(z) \le B(m+3k) - \alpha(N-3k).$$

Combining this with (5.3) bounds the $D_n$-length of the middle path by

$$\begin{aligned}\operatorname{len}_{D_n}(\gamma_P) &\le 2\exp[(\log(11/8) - \psi\alpha)N + \psi Bm + 3\psi(B+\alpha)k] \\ &= 2\exp[-\beta n + (\beta + \psi B)m + 3\psi(B+\alpha)k] \\ &\le 2\exp[-(\log 2 + 3\kappa/4)n + 3\psi(B+\alpha)k] \le 2^{-n}\end{aligned} \tag{5.7}$$

for all sufficiently large $n$. Concatenating the two attachment paths and the middle bridge inside $\mathcal{C}$ gives, at the level of infima,

$$D_n(0, e_1; \mathcal{C}) \le 2T_n + 2^{-n}$$

on the event in (5.6). Exact geodesics or measurable minimizing paths are not needed. Since the event has probability strictly larger than $1/2$, we have proved the deterministic recursion

$$\boxed{a_n \le 2e^{C\sqrt{n}} 2^{-\lfloor\theta n\rfloor} a_{\,n-\lfloor\theta n\rfloor} + 2^{-n}} \tag{5.8}$$

for all sufficiently large $n$.

## 6 Iteration, lower bound, and the exponent

Define $b_n = 2^n a_n$ and $f_n = 1 + b_n$. Equation (5.8) gives

$$b_n \le 2e^{C\sqrt{n}} b_{\,n-\lfloor\theta n\rfloor} + 1, \qquad f_n \le e^{C'\sqrt{n}} f_{\,n-\lfloor\theta n\rfloor}$$

for all sufficiently large $n$. Increase the fixed threshold $n_*$ so that $n_* \ge 2/\theta$. Starting from $n_0 = n$, successively set

$$n_{j+1} = n_j - \lfloor\theta n_j\rfloor$$

while $n_j \ge n_*$. Then

$$n_{j+1} \le (1-\theta/2)n_j, \qquad \sum_j \sqrt{n_j} \le \frac{\sqrt{n}}{1-\sqrt{1-\theta/2}}.$$

Iterating the recursion until the index is below $n_*$ gives

$$f_n \le \Big(\max_{1\le \ell<n_*} f_\ell\Big)\exp\!\left(\frac{C'\sqrt{n}}{1-\sqrt{1-\theta/2}}\right).$$

All of the finitely many initial medians are positive and finite. Enlarging the constant proves

$$a_n \le 2^{-n}e^{C\sqrt{n}} \qquad (n\ge 1). \tag{6.1}$$

For the matching lower bound, every continuous path from 0 to $e_1$ must leave the Euclidean ball $B_{2^{-n}}(0)$. The initial portion up to its first exit has Euclidean length at least $2^{-n}$. Therefore, pathwise,

$$D_n(0,e_1) \ge 2^{-n}\exp\!\left(\psi \inf_{|z|_2\le 2^{-n}} h_n(z)\right). \tag{6.2}$$

Lemma 2.2(ii), with $m=n$ and $R=1$, shows that the infimum is at least $-C\sqrt{n}$ with probability at least 0.99. Thus

$$\operatorname{Med} D_n(0,e_1) \ge 2^{-n}e^{-C\sqrt{n}}. \tag{6.3}$$

Since unrestricted distance is at most the internal distance, (6.1) and (6.3) prove (1.4).

In particular,

$$\log \operatorname{Med} D_n(0,e_1) = -n\log 2 + O(\sqrt{n}).$$

Comparing with the defining identity

$$\log \operatorname{Med} D_n(0,e_1) = (-1+\psi Q_d(\psi))n\log 2 + o(n)$$

gives $Q_d(\psi)=0$. In fact, the proof has established the exponential rate directly, without using prior existence of $Q_d(\psi)$.

For fixed $x\ne y$, let $S(u)=x+|y-x|_2Ru$ with $Re_1=(y-x)/|y-x|_2$. The dilation argument in Lemma 2.2(iv) works for this fixed scale as well: the difference consists of two bands of fixed width $|\log_2|y-x|_2|$, whose supremum on $\mathcal{C}$ is $O_{x,y}(\sqrt{n})$ in every fixed quantile. This transfers the upper bound to the internal distance in $S\mathcal{C}$, and hence to the unrestricted distance. The local escape argument around $x$ gives the lower bound for all sufficiently large $n$. Consequently

$$\log \operatorname{Med} D_n(x,y) = -n\log 2 + O_{x,y}(\sqrt{n}).$$

Lemma 2.1 gives the same bound for any fixed quantile. For every fixed $\varepsilon>0$, the probabilities

$$\mathbb{P}\big[|\log D_n(x,y) - \log \operatorname{Med} D_n(x,y)| > \varepsilon n\big]$$

are summable in $n$. Borel–Cantelli, first for positive rational $\varepsilon$, proves (1.5) and completes Theorem 1.1.

## 7 The set-to-set exponent in the same regime

The preceding point-to-point proof did not use the set-to-set exponent. To identify the two exponents jointly, now take as input the existence theorem supplied in [2, Theorem 1.1]:

$$\operatorname{Med} D_n(K_1,K_2) = 2^{(-1+\psi\widetilde{Q}_d(\psi))n+o(n)} \tag{7.1}$$

for disjoint compact sets with nonempty interiors.

Fix $M = 10$ in Lemma 3.1 and let

$$K_1 = \overline{B}_2(0), \qquad K_2 = \overline{B}_2(20e_1).$$

All candidate paths lie in a fixed compact rectangle $V$. Since $k$ is fixed and $h_{3k}$ is continuous, choose deterministic $L < \infty$ with

$$\mathbb{P}\left[\sup_V h_{3k} \le L\right] \ge 0.99.$$

This is a decreasing event. Lemma 4.1, now with the same coarse-field event for each candidate, gives probability at least $0.99^2/(5/4) > 1/2$ for a good bridge on which this bound also holds. On that event its metric length is at most

$$40(11/8)^n \exp\big(\psi L - \psi\alpha(n-3k)\big) = C\exp\big((\log(11/8) - \psi\alpha)n\big).$$

Thus (7.1) implies

$$\widetilde{Q}_d(\psi) \le \frac{\log(11/4) - \psi\alpha}{\psi \log 2} < 0. \tag{7.2}$$

Combining this with Theorem 1.1, throughout (1.3) we obtain

$$\boxed{\widetilde{Q}_d(\psi) < 0 = Q_d(\psi).}$$

The proof does not assert that the scalar inequality $\widetilde{Q}_d(\psi) < 0$ by itself forces $Q_d(\psi) = 0$ in every dimension or parameter range. It uses the monotone weighted bridge construction, uniformly in its macroscopic length, to resolve the endpoint attachment problem in the high-dimensional regime.

## References

[1] J. Ding, E. Gwynne, and Z. Zhuang. *Percolation of thick points of the log-correlated Gaussian field in high dimensions.* arXiv:2410.22075, version 2 (February 2026). The input used here is Definition 3.13, Equation (3.16), Equations (3.43)–(3.46), and the bridge modification in the proof of Proposition 3.19. `https://arxiv.org/abs/2410.22075v2`.

[2] *Existence of the set-to-set distance exponent: In the white-noise setup of Ding–Gwynne–Zhuang.* User-supplied note dated October 24, 2026, filename `set_to_set_exponent(1).pdf`, Theorem 1.1. Used only in Section 7.